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Analyzing Different Transfer Functions for Microbiological Transfer Learning

Framework:

One of the most well-studied microbes in the biological literature is *Escherichia coli*, with over 970,000 genomes available in public databases (Ussery, 2024). However, over 99% of living microbes are unknown in science (Jiao et al., 2021). With the few microbes which are known, many including *S. aureus* are less extensively studied. An approach to attempt to understand these other microbes better is transfer learning, where a machine learning model is first pre-trained on a large dataset (such as the one for *E. coli*) and then fine-tuned with one of these microbes' datasets. However, fine-tuning such a model involves a way to map features from the original, larger dataset to features in the smaller dataset. The means of this mapping is through a transfer function, of which there are many different methods.

Our research attempts to evaluate different transfer functions to align *E. coli* features to *S. aureus* features. The different methods that can be used as transfer functions include *k*-mer mapping with tools such as Salmon (Patro et al., 2015), nucleotide-based alignment with BLAST (Healy, 2007), and a higher level protein-based alignment with models such as AlphaFold (Jumper et al., 2021). Different transfer functions can have varying effects on model performance, and we aim to evaluate these functions on which ones assist in training a better model with *E. coli* providing the pre-training dataset and *S. aureus* providing the fine-tuning dataset.

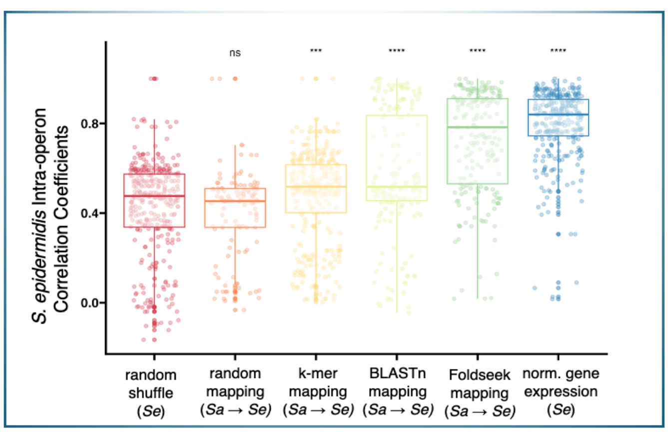

Figure 1 In this side-by-side box plot, *S. aureus* was mapped to *S. epidermidis* through different transfer learning functions. The first two box plots on the left represent negative control groups, either with random mapping or shuffling the genome. The three box-plots afterward represent *k*-mer mapping, nucleotide alignment, and protein-based mapping, from left to right. The last box plot is a positive control group, where *S. epidermidis* was mapped to its own genome. The three functions each had different effects on the model, as seen with the varying correlation coefficients on the vertical axis.

The side-by-side box plot in Figure 1 resulted from a previous analysis on different transfer functions and mapping *S. aureus* to *S. epidermidis*. In this project, we take this research as a precedent for transfer learning and build off of it to map an *E. coli* dataset to a *S. aureus* dataset. The *E. coli* dataset we will use is ten times larger than the pre-training dataset used in this previous analysis, with over 20,000 samples total. Also, *E. coli* is more distantly related to *S. aureus* in comparison to this past research, which might reveal new insights about transfer learning performance and phylogenetic distance.

Goals:

This project aims to answer the following research **question**: which transfer function produces the best transfer learning performance when fine-tuning a machine learning model pre-trained on *E. coli* with *S. aureus* data?

In the process of answering our research question, we will learn whether different transfer functions behave differently when fine-tuning on the same dataset. We will also examine whether the phylogenetic distance between the original pre-training microbe and the lesser-studied microbe has an impact on the model performance. Model performance will be assessed using both the model's loss function and intra-operon correlations (as in Figure 1), acting as a proxy to capture biological pathways. Overall, we will also look at whether fine-tuning actually benefits the pre-trained model by increasing performance or detriments it by simply adding random noise.

We introduce the main **hypothesis** in regards to our research question: Utilizing *k*-mer mapping as the transfer function will optimize the transfer learning performance of the model, since it works better with more distantly related organisms unlike nucleotide-based BLAST alignment and predictive protein-level mapping, which might only be best for closely related organisms.

Evaluating different transfer functions for model fine-tuning is a specific part of asking a broader question: will fine-tuning actually benefit or disadvantage the model? In one part, introducing a smaller dataset could help with mapping genes and sequence alignment due to similarities in the pre-training and fine-tuning datasets. However, the new dataset could simply introduce noise into the model, weakening rather than strengthening it. This research hopes to address this higher-level question and investigate deeper into what characterizes different functions and their corresponding models.

The anticipated outcomes of this project include building different neural networks on different transfer functions and analyzing their performance, comparing them, and seeing the overall effect of the phylogenetic distance between the pre-training and fine-tuning microbes. This will build off of previous work done with mapping *S. aureus* to *S. epidermidis* with different functions (Tan et al., 2017), and analysis will be used to make conclusions and extract meaningful insights about transfer learning in microbiology.

Impact:

The field of microbiology has vast potential to grow with the study of many unknown and lesser known microorganisms, and one of the main reasons they are less studied is because so little data is available about these microbes. The more that scientists and researchers know about these microbes, the more research that can be done about applications of these microbes to solve or be better prepared for real-world problems.

We aim to see how using transfer learning, where we harness the power of a large dataset combined with a smaller but phylogenetically related dataset, can help the field understand lesser-known microbes better. If we find that this process yields statistically significant results, then future research can focus on utilizing transfer learning and neural networks to study other microbes based on an already well-studied microbe. The world of "microbial dark matter" still remains mysterious with so many unknown microbes, and we hope that our research will take a step towards a better understanding of this interesting but obscure world (Jiao et al., 2021).

Plan:

The main steps for this project can be split into four main stages: pre-training, evaluation, fine-tuning, and then a final thorough evaluation. The first stage will involve training a Denoising Autoencoder (a type of neural network) from scratch using a large *E. coli* dataset (Tan et al., 2017). To do this, we expect to use the Keras Python package (https://pypi.org/project/keras/) and build off of other tools already developed for *S. aureus* and *S. epidermidis* (https://github.com/georgiadoing/seqADAGE). The second stage, or the first evaluation, will be examining this pre-trained model as a control and assessing its performance, so that comparison can be made to the same model after fine-tuning.

The third part will be fine-tuning the trained model with transfer learning by utilizing different transfer functions discussed earlier. This will be divided into different parts with each using a different transfer function to map *E. coli* features to *S. aureus* features before fine-tuning. Then, the actual fine-tuning will happen, where the model is trained on the smaller dataset. This will produce different neural networks that were trained with different transfer functions for evaluation.

The final part of the project will be evaluation and interpretation. At this stage, we will compare the performance metrics of the pre-trained model only to the transfer learned model to find significant patterns. This evaluation stage will compare both models for *S. aureus* using AureoWiki (https://aureowiki.med.uni-greifswald.de/) to obtain pre-determined genes in existing databases as well as EcoCyc (https://ecocyc.org/) to assess if the model has learned *E. coli* pathways. From this, we will draw correlations and conclusions about each transfer function, its overall impact on a model's quality, and how it relates to the phylogenetic distance from *S. aureus* to *E. coli*. This process will involve pathway enrichment analysis.

Personal Goals:

I believe that this project supports my long-term career goals by allowing me to work extensively with machine learning and genomics. I am especially interested in using different types of machine learning methods in the scientific field to extract newer insights for real-world problems. Especially in using transfer learning in microbiology, there are many challenges since the feature spaces don't match for *E. coli* and the lesser-studied microbe. The genes are different, the gene names don't match, and genomes vary based on the organism. Microbial transfer learning presents a different real-world problem compared to traditional transfer learning, where models trained on images of cats can be fine-tuned on dogs since the feature spaces are essentially the same—pixels with *x* and *y* coordinates.

Through this research project, I will gain experience with working with neural networks and transfer learning, evaluating different models and how their performance is shaped by input types, and how to conduct independent research and test statistical hypotheses to produce meaningful results. In terms of my professional career, I hope that this project aligns with my long-term goal of applying machine learning techniques in a future career in industry as a machine learning engineer or a data scientist. Overall, I believe that this project will help me develop creative problem-solving skills and experience with machine learning which will help me in pursuing a career in industry. I'm grateful to Professor Georgia Doing for agreeing to advise this project and the Committee on Undergraduate Research for reviewing my research proposal.

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